| Question | | | Answer | Marks | AOs | Guidance | |
|---|---|---|---|---|---|---|---|
| 13 | (a) | | $\mathbf{r} = \int (\mathbf{v})\, dt = \int \left(3t^2\mathbf{i} + 7\mathbf{j}\right) dt$<br>$= t^3\mathbf{i} + 7t\mathbf{j} + \mathbf{c}$ | M1* | 3.4 | Attempt to integrate both components; condone missing +**c** | |
| | | | When $t = 0$, $\mathbf{r} = -\mathbf{i} + 2\mathbf{j} = \mathbf{c}$ | M1 dep* | 1.1b | Using initial conditions | |
| | | | So $\mathbf{r} = \left(t^3 - 1\right)\mathbf{i} + \left(7t + 2\right)\mathbf{j}$ | M1 dep* | 3.1b | Using $t = 2$ to find position vector or values for $x$ and $y$ | |
| | | | When $t = 2$,<br>$\mathbf{r} = \left(2^3 - 1\right)\mathbf{i} + \left(7 \times 2 + 2\right)\mathbf{j} = 7\mathbf{i} + 16\mathbf{j}$ | A1 | 1.1b | Accept vector form or two clear components. | |
| | | | distance = $\sqrt{7^2 + 16^2} = \sqrt{305}$<br>distance = 17.5 m | M1<br>A1<br>[6] | 3.1b<br>3.2a | Using Pythagoras<br>FT their components | |
| 13 | (b) | | using $x = t^3 - 1$, $y = 7t + 2$ | M1 | 3.1a | Extracting equations for $x$ and $y$ from their displacement vector | |
| | | | Substitute $t = \dfrac{y-2}{7}$ into equation for $x$ | M1 dep | 1.1b | Attempt to eliminate $t$ | Equivalent form $y = 7\left(x+1\right)^{\frac{1}{3}} + 2$ for M1M1A0 |
| | | | $x = \left(\dfrac{y-2}{7}\right)^3 - 1$ **AG** | A1<br>[3] | 1.1b | cao | |
| 13 | (c) | | $\mathbf{a} = \dfrac{d\mathbf{v}}{dt} = 6t\mathbf{i}$ | M1* | 3.1b | Must be vector acceleration | |
| | | | When $t = 2$, $\mathbf{a} = 12\mathbf{i}$ | M1* | 3.4 | Evaluating when $t = 2$ | $a$ =12 is sufficient here |
| | | | The force must be in that direction, so $\mathbf{F} = 48\mathbf{i} = m\mathbf{a}$ | M1* | 3.1b | Newton's second law in vector form, or in $x$-direction only | If their **a** has two non-zero components, allow for dividing 48 by the magnitude of their **a** |
| | | | So $m = 4$ kg | A1 dep*<br>[4] | 1.1b | cao | |